KENTARO ISHII, PH.D

University of Washington
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Current status

Center for the Neurobiology of Addiction, Pain, and Emotion, Department of Anesthesiology and Pain Medicine, Department of Pharmacology, University of Washington
in the laboratory of Prof. Garret Stuber **2019-present**

Education

Ph. D Department of Applied Biological Chemistry, Graduate School of Agricultural and Life Sciences, The University of Tokyo **2019**

- Dissertation title: Analysis of the neural basis that controls odor and pheromone evoked innate behavior

M.S. Department of Applied Biological Chemistry, Graduate School of Agricultural and Life Sciences, The University of Tokyo **2016**

B.S. Department of Applied Biological Chemistry and Technology, The University of Tokyo **2014**

Experience

Acting instructor, Advisor: Prof. Garret Stuber, University of Washington **2024-**

- Characterized morphine responsive neurons in the mouse brain
- Constructing pre-processing pipeline for whole brain imaging

Postdoctoral research scholar, Advisor: Prof. Garret Stuber, University of Washington **2019-2024**

- Characterized estradiol sensitive neurons in the paraventricular hypothalamus.
- Identified neurons in the medial preoptic area which regulates sexual motivation

The University of Tokyo, Graduate School of Agricultural and Life Sciences, Department of Applied Biological Chemistry
Graduate Student (Doctor-Course) in the laboratory of Prof. Kazushige Touhara 2014-2019

- Identified neural circuit conveying a juvenile pheromone information in mice.
- Identified neural circuit conveying sex pheromone information in mice.
- Assisted in one under graduate-level training course.

The University of Tokyo, Department of Applied Biological Chemistry and Technology
Under Graduate Student in the laboratory of Prof. Kazushige Touhara 2013-2014

- Identified the type of hypothalamic neuron responding to sex pheromone stimuli in mice.

Awards

Dean award of the Department of Applied Biological Chemistry (Doctor-course)	2019
Young Investigator Travel Award of the 17th International Symposium on Olfaction and Taste	2016
Poster award of 16 th Univesity of Tokyo Biology Symposium	2016
Dean award of the Department of Applied Biological Chemistry (Master-course)	2016

Research Funding and Fellowship

Overseas Post-Doctoral Fellowship from Uehara Memorial Foundation		
	4,100,000 JPY per year	2021-2022
Overseas Research Fellowship from JSPS	55,700 USD per year	2019-2021
Research Fellowship for Young Scientists from JSPS	2,400,000 JPY per year	2016-2019

Presentations and Publications

Publication

1. "A Labeled-Line Neural Circuit for Pheromone-Mediated Sexual Behaviors in Mice."

Ishii KK*, Osakada T*, Mori H, Miyasaka N, Yoshihara Y, Miyamichi K, Touhara K. *Neuron* **95,** 123–137.e8 (2017). * equal contribution.

2. "Sexual rejection in female mice via a vomeronasal receptor-triggered limbic circuit."

Osakada, T.*, **Ishii, K.K.***, Mori, H*., Eguchi, R., Ferrero, D.M., Yoshihara, Y., Liberles, S.D., Miyamichi, K., and Touhara, K. (2018). *Nat. Commun*. 9, 4463. * equal contribution.

3. "Hypothalamic Neuronal Circuits Regulating Hunger-Induced Taste Modification"

Fu, O., Iwai, Y., Narukawa, M., Ishikawa, A. W., **Ishii, K. K.**, Murata, K., Yoshimura, Y., Touhara, K., Misaka, T., Minokoshi, Y., & Nakajima, K. I. (2019). *Nat. Commun.* 10 (1), 4560.

4. "Hemoglobin in the blood acts as a chemosensory signal via the mouse vomeronasal system"

Osakada T., Abe T., Itakura T., Mori H., **Ishii K. K.**, Eguchi R., Murata K., Saito K., Haga-Yamanaka S., Kimoto H., Yoshihara Y., Miyamichi K. & Touhara K. (2022). Nat. Commun. 13 (1), 1-14

5. "A single vomeronasal receptor promotes intermale aggression through dedicated hypothalamic neurons"

Itakura T., Murata K., Miyamichi K., **Ishii K. K.**,, Yoshihara Y., Touhara K. (2022). *Neuron 110*, 2455-2469.e8.

6. "Post-ejaculatory Inhibition of Female Sexual Drive via Heterogeneous Neuronal Ensembles in the Medial Preoptic Area"

Ishii, K.K., Hashikawa, K., Chea, J., Yin, S., Fox, R.E., Kan, S., Shah, M., Zhou, Z.C., Navarrete, J., Murry, A.D., Szelenyi, E.R., Golden, S.A., and Stuber, G.D. (2025). eLife 12, RP91765.

7. "Opioid-driven disruption of the septal complex reveals a role for neurotensin-expressing neurons in withdrawal"

Simon, R.C., Fleming, W.T., Briones, B.A., Trzeciak, M., Senthilkumar, P., **Ishii, K.K.**, Hjort, M.M., Martin, M.M., Hashikawa, K., Sanders, A.D., Golden, S.A., and Stuber, G.D. (2025). *Neuron*, *113*, 2325 - 2343.e9.

8. "Esr1-Dependent Signaling and Transcriptional Maturation in the Medial Preoptic Area of the Hypothalamus Shapes the Development of Mating Behavior during Adolescence"

Hashikawa, K., Hashikawa, Y., Briones, B., **Ishii, K.**, Liu, Y., Rossi, M.A., Basiri, M.L., Chen, J.Y., Ahmad, O.R., Mukundan, R.V., Johnston, N.L., Simon, R.C., Soetedjo, J.C., Siputro, J.R., McHenry, J.A., Palmiter, R.D., Rubinow, D.R., Zweifel, L.S., and Stuber, G.D. (2025). *eLife* **14**, RP106347.

9. "Hierarchical and Spatial Mapping of Whole-Brain c-Fos Activity Reveals Distinct Opioid and Withdrawal Neuronal Ensembles"

Ishii, K.K., Zhang, Y., Ren, L., De Lecea, A., Bakshi, S., Haygood, A., Chujo, S., Stevenson, H.P., Schwartz, E., Zhou, Z.C., Szelenyi, E.R., Golden, S.A., Kessler, D., and Stuber, G.D. (2025). *bioRxiv* 2025.09.21.677198.

Presentation

1. "Processing of pheromone information within the MeA→VMH circuit"
   Kentaro Ishii, Kazunari Miyamichi, Kazushige Touhara. **The 2nd Chemosensory and Behavior workshop** 2014, Oral presentation
2. "Dissecting the MeA→VMH circuit responsible for sex pheromone-mediated behavior"
   Kentaro Ishii, Kazunari Miyamichi, Kazushige Touhara. **The 38th Annual Meeting of the Japan Neuroscience Society** 2015, Poster presentation.
3. "Dissecting the neural circuit responsible for sex pheromone mediated behavior"
   Kentaro Ishii, Kazunari Miyamichi, Kazushige Touhara. **16 th Univesity of Tokyo Biology Symposium.** Poster presentation.
4. "Dissecting the neural circuit responsible for sex pheromone mediated behavior"
   Kentaro Ishii, Kazunari Miyamichi, Kazushige Touhara. **17th International Symposium on Olfaction and Taste 2016.** Poster presentation.
5. "An allocated neural circuit in the amygdala-hypothalamus axis for sex pheromone-mediated behavior in female mice"
   Kentaro Ishii, Kazunari Miyamichi, Kazushige Touhara. **"Optogenetic Approaches to Understanding Neural Circuits & Behavior" Gordon Research Conference 2016.** Poster presentation.
6. "Positive and negative control of female sexual behavior"
   Kentaro Ishii, Takuya Osakada, Hiromi Mori, Kazunari Miyamichi, Kazushige Touhara. **Optogenetics Research Conference Japan 2017.** Poster presentation.
7. "Neural circuitry for positive and negative modulation of pheromone-mediated female sexual behavior"
   Kentaro Ishii, Takuya Osakada, Hiromi Mori, Kazunari Miyamichi, Kazushige Touhara. **The seventh Cold Spring Harbor conference on Neuronal Circuits 2018.** Poster presentation
8. "Representation of female sexual behavior in the medial preoptic area"
   Kentaro K Ishii, Garret Stuber. **NAPE seminar series 2022, University of Washington.** Oral presentation.
9. "Dissecting sensory and motivational signals in the MPOA during mating in female mice"
   Kentaro K Ishii, Adam Gordon-Fennell, Koichi Hashikawa, Jane Chea, Shihan Yin, Rebecca Erin Fox, Garret Stuber. **"Hypothalamus" Gordon Research Conference 2022.** Poster presentation.
10. "Inhibitory control of sexual motivation by the medial preoptic area"
    Kentaro K Ishii, Garret Stuber,
    Kobe symposium 2023. 3/9/2023. Oral presentation.
11. "Inhibitory control of sexual motivation by the medial preoptic area"
    Kentaro K Ishii, Garret Stuber.
    Invited guest at Kobe University. 3/23/2023. Oral presentation.
12. "Inhibitory control of sexual motivation by the medial preoptic area"
    Kentaro K Ishii, Garret Stuber.
    Invited speaker at "The 5th future of neurocircuitry research workshop". 3/24/2023. Oral presentation.
13. "Brain Wide neuronal ensembles engaged by opioids"
    Kentaro K Ishii, Garret Stuber.
    Invited speaker at "2024 Department of Pharmacology Retreat Symposium on Molecular Pharmacology" 9/20/2024. Oral presentation.
14. "Brain wide neuronal ensembles engaged by opioid"
    Kentaro K Ishii, Garret Stuber.
    Invited guest at Kobe University. 11/18/2024. Oral presentation.
15. "Inhibitory control of sexual motivation by the medial preoptic area"
    Kentaro K Ishii, Garret Stuber.
    Invited guest at Tokyo University. 11/20/2024. Oral presentation.
16. "Whole brain analysis of opioid responding ensemble using improved multiple comparison correction algorithm "
    Kentaro K Ishii, Garret Stuber.
    Invited guest at Juntendo University. 11/21/2024. Oral presentation.
17. "Brain Wide Neural Ensemble Engaged by Opioid"
    Kentaro K. Ishii
    "2025 SPEAR conference", Friday Harbor Labs, University of Washington. 5/11/2025. Oral presentation.
18. "Neural substrate for post-ejaculatory inhibition of female sexual drive"
    Kentaro K Ishii, Garret Stuber.

Guest speaker at "Neuroscience seminar series", University of Washington. 6/2/2025. Oral presentation.

19. "Brain-wide mapping of neural ensembles during morphine addiction using a novel error rate and testing strategy"
Kentaro K Ishii, Daniel Kessler, Garret D. Stuber
"2025 CNCM conference The Changing Brain", UC Irvine. 8/18/2025. Poster presentation.

Reviews, Commentaries and Book chapters

1. "The neural circuit determining the valence of odors" (Review)
Kentaro Ishii, Kazushige Touhara *Journal of Japan Association on Odor and Environment* Vol. 45, No.4, 2015 (in Japanese)
2. "Neural circuits regulating sexual behaviors via the olfactory system in mice." (Review)
Ishii, K.K., and Touhara, K. (2019). *Neurosci. Res*.
3. "Encyclopedia of Living Organisms, Odors, and Pheromones" (Book chapter)
Kazushige Touhara *et al.* (2025). Chapter 2.5.5. *Asakura publishing co.*

Book Chapters

1. "Encyclopedia of Living Things, Scents, and Pheromones"
Ishii, K. "The amygdala". In K. Touhara, Y. Niimura, Y. Yoshihara, M. Yokosuka, T. Kikusui, & M. Okamoto (Eds.), *Encyclopedia of animals and odors & pheromones* (Chap. 2.5.5, pp. 118-121). Asakura Publishing.

Workshop

1. "IMAGING &GENETICS WORKSHOP 2021"
Charles Zhou, Abbie Schindler, Jane Chen, Kentaro Ishii, Sean Piantadosi, Adam Gordon-Fennell, Raaj Gowrishankar, Madelyn Hjort, Lusine Gomtsian
"Virus injection, GRIN lens, and headcap implantation surgery" University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing and launching the workshop.
2. "IMAGING &GENETICS WORKSHOP 2022"
Charles Zhou, Abbie Schindler, Jane Chen, Kentaro Ishii, Sean Piantadosi, Adam Gordon-Fennell, Raaj Gowrishankar, Madelyn Hjort, Lusine Gomtsian
"Virus injection, GRIN lens, and headcap implantation surgery" University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing the workshop.
3. "IMAGING &GENETICS WORKSHOP 2023"
Charles Zhou, Abbie Schindler, Jane Chen, Kentaro Ishii, Sean Piantadosi, Adam Gordon-Fennell, Raaj Gowrishankar, Madelyn Hjort, Lusine Gomtsian
"Virus injection, GRIN lens, and headcap implantation surgery" University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing the workshop.
4. "LIGHT SHEET IMAGING WORKSHOP 2024"
Charles Zhou, Kentaro Ishii, Eric Szelenyi, Jingyi Chen, Jovana Navarrete, Sam Golden, Lusine Gomtsian
"Demonstration of tissue clearing using the LifeCanvas SmartBatch system and preprocessing pipeline using high performance cluster computer" University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing the workshop.

Mentorship

Stuber lab, University of Washington

Year	Name	Position	Current Position
2019-2022	Ned (Suyang) Kan	Undergraduate Researcher	Ph.D. student, Rebecca Yang lab, Duke University
2022-2023	Shihan Yin	Undergraduate Researcher	Ph.D. student, Colin Palmer lab, National University of Singapore
2023	Ekayana Sethi	Rotation Ph.D.	Ph.D. student, Larry Zweifel lab, University of Washington
2022-2024	Jane Chea	Undergraduate Researcher	Ph.D. student, NYU Grossman School of Medicine
2022-2024	Rebecca E. Fox **(Jeremy Clark Memorial Fellow)**	Undergraduate Researcher	Research technician at Seattle Children's Applying for medical school
2024	Yizhe Zhang	Rotation Ph.D.	Ph.D. student, Astra S. Bryant lab, University of Washington
2024-2025	Meha Shah **(Jeremy Clark Memorial Fellow)**	Undergraduate Researcher	N/A
2025	Sakura Chujo	International internship	Ph.D. student, Yukiko Gotoh lab, University of Tokyo
2024-Present	Srishti Bakshi	Undergraduate Researcher	N/A
2024-Present	Ariyanna Haygood **(NAPE Undergrad. Research Program)**	Undergraduate Researcher	N/A

Touhara lab, University of Tokyo

Year	Name	Position	Current Position
2016-2018	Koki Emori	Ph. D. Student	N/A
2017-2018	Miki Matsumoto	Undergraduate Researcher	N/A
2017-2019	Takumi Itakura	Ph. D. Student	Postdoc Scholar, David Anderson lab, California Institute of Technology

References

Garret Stuber, Ph.D., Professor
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